DATA SCIENCE PROJECT REPORT

AIR QUALITY ANALYSIS USING PYTHON

Submitted by

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CERTIFICATE

This is to certify that Mauli Mahiya bearing Registration No. 12408949 has successfully completed the project titled 'Air Quality Analysis using Python' under my supervision. The work is original.

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DECLARATION

I, **Mauli Mahiya**, a student of **B.Tech Computer Science and Engineering** under the Discipline of CSE at **Lovely Professional University, Punjab**, hereby solemnly declare that the project report entitled **"Air Quality Analysis using Python"** submitted in partial fulfillment of the requirements for the award of the degree is a record of my own original work carried out by me during the academic session.

I affirm that this project work has been undertaken by me under the guidance and supervision of my faculty mentor. The work presented in this report has not been copied from any source, and wherever external sources of information, data, or references have been used, they have been properly acknowledged and cited in accordance with standard academic practices.

I further declare that this project has not been submitted either in whole or in part to any other university, institution, or organization for the award of any degree, diploma, or certification. The analysis, interpretations, and conclusions drawn in this report are entirely based on my own understanding and implementation of the concepts learned during the course.

I take full responsibility for the authenticity, originality, and accuracy of the work presented in this report. I understand that any violation of academic integrity, if found, may lead to disciplinary action as per the rules and regulations of the university.

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1. INTRODUCTION

Air pollution is one of the most severe environmental challenges faced by the world today, affecting not only the ecosystem but also human health and economic development. With the rapid growth of industrialization, urbanization, and population, the quality of air has significantly deteriorated over the past few decades. Major cities, especially in developing countries like India, are experiencing alarming levels of air pollution due to increasing vehicular emissions, industrial activities, construction work, burning of fossil fuels, and other human activities. As a result, maintaining a clean and healthy environment has become a critical concern for governments, environmental organizations, and society as a whole.

Air pollution consists of a mixture of harmful substances known as **air pollutants**, which include **Particulate Matter (PM2.5 and PM10), Nitrogen Dioxide (NO_2), Sulfur Dioxide (SO_2), Carbon Monoxide (CO), and Ozone (O_3)**. Among these, **PM2.5 and PM10** are considered the most hazardous pollutants because they are extremely fine particles that can easily penetrate deep into the lungs and bloodstream. Exposure to these pollutants can lead to serious health issues such as respiratory infections, asthma, lung cancer, cardiovascular diseases, and even premature death. In addition to health impacts, air pollution also contributes to environmental problems such as global warming, acid rain, and climate change.

Due to these serious consequences, monitoring and analyzing air quality has become an essential task. Governments and environmental agencies collect large amounts of air quality data on a regular basis. However, raw data alone is often difficult to interpret and does not provide meaningful insights without proper analysis. Therefore, there is a strong need for applying **data science techniques**, including data preprocessing, visualization, statistical analysis, and machine learning, to transform raw data into useful information.

This project, titled **"Air Quality Analysis using Python,"** focuses on analyzing air pollution data using modern data science tools and techniques. The dataset used in this project contains approximately **12,000 records**, including attributes such as city, date, and levels of various pollutants. The project begins with **data preprocessing**, which is a crucial step in data analysis. In this stage, the dataset is cleaned by handling missing values, converting date formats, and creating new features such as **month, day, hour, and time categories (day and night)**. These transformations help in improving the quality of data and make it suitable for further analysis.

After preprocessing, **Exploratory Data Analysis (EDA)** is performed to understand the structure, distribution, and relationships within the dataset. Various visualization techniques such as **bar charts, pie charts, box plots, scatter plots, and heatmaps** are used to represent the data in a clear and intuitive manner. These visualizations make it easier to identify trends, compare pollution levels across different cities, and understand the behavior of different pollutants.

The project includes several important analyses such as:

- **City-wise pollution comparison** to identify the most polluted regions
- **Pollutant distribution analysis** to determine the most critical pollutants
- **Day vs Night analysis** to study variation in pollution over time
- **Weekday vs Weekend analysis** to observe behavioral patterns in pollution levels
- **Correlation analysis** to understand relationships between different pollutants

These analyses provide a comprehensive understanding of how pollution varies across different conditions and help in identifying key factors affecting air quality.

In addition to descriptive analysis, this project also incorporates **Machine Learning techniques**, specifically **Linear Regression**, to predict **PM2.5 levels using PM10 values**. The objective is to examine whether there exists a strong relationship between these two pollutants and whether one can be used to predict the other. The model is evaluated using performance metrics such as **R^2 Score, Mean Squared Error (MSE), and Mean Absolute Error (MAE)**. This step highlights both the potential and limitations of simple machine learning models when applied to complex environmental data.

The results obtained from this project indicate that pollution levels across all cities are significantly higher than the safe limits prescribed by the **World Health Organization (WHO)**. It is also observed that there is **weak correlation between certain pollutants**, suggesting that air pollution is influenced by multiple independent factors rather than a single source. Furthermore, the machine learning model shows **low prediction accuracy**, which indicates that more advanced models and additional features are required for better prediction.

Overall, this project demonstrates the practical application of **data science, data visualization, and machine learning techniques** in solving real-world environmental problems. It emphasizes the importance of analyzing environmental data to gain insights, support decision-making, and create awareness about pollution. The knowledge and techniques used in this project can be further extended to develop **real-time monitoring systems, advanced predictive models, and intelligent dashboards** that can assist in controlling and reducing air pollution.

In conclusion, this project not only enhances the understanding of air quality data but also highlights the importance of technology in addressing environmental challenges. It provides a strong foundation for future research and development in the field of environmental data analysis and smart pollution management systems.

SCOPE OF ANALYSIS

The scope of this project is centered around the comprehensive analysis of air quality data using data science and machine learning techniques to extract meaningful insights and understand pollution patterns across different cities and time periods. With the increasing concern over environmental degradation and health risks caused by air pollution, analyzing such data becomes crucial for awareness and decision-making.

This project focuses on a dataset containing approximately **12,000 records**, which includes various attributes such as **city, date, and pollutant levels** like **PM2.5, PM10, NO_2, SO_2, CO, and O_3**. The scope of the analysis involves multiple stages, including **data preprocessing, exploratory data analysis (EDA), visualization, statistical analysis, and predictive modeling**.

2.1 Data Preprocessing Scope

The first step within the scope involves **data preprocessing**, which ensures that the dataset is clean, structured, and ready for analysis. This includes:

- Converting the **date column into proper datetime format**
- Extracting useful features such as **month, day, and hour**
- Creating new attributes like **Day/Night classification**
- Categorizing data into **Weekday and Weekend**

These preprocessing steps enhance the dataset by adding meaningful features that allow deeper analysis of pollution patterns.

2.2 City-wise Analysis

A major part of the scope includes **city-wise comparison of pollution levels**. In this analysis:

- Average pollution levels are calculated for each city
- Cities are ranked based on **PM2.5 levels**
- The most polluted and least polluted cities are identified

This helps in understanding how pollution varies geographically and highlights critical regions where pollution control measures are required.

2.3 Pollutant-wise Analysis

The project also focuses on analyzing different pollutants individually to determine their impact on overall air quality. The pollutants considered include:

- **PM2.5 (Fine Particulate Matter)**
- **PM10 (Coarse Particulate Matter)**
- **NO_2 (Nitrogen Dioxide)**
- **SO_2 (Sulfur Dioxide)**
- **CO (Carbon Monoxide)**
- **O_3 (Ozone)**

The scope includes:

- Comparing average levels of each pollutant
- Identifying the **most critical pollutant**
- Studying how each pollutant behaves across different cities

This analysis provides insights into which pollutants contribute the most to air pollution.

2.4 Time-based Analysis

Another important aspect of the scope is analyzing pollution based on time. This includes:

2.4.1 Day vs Night Analysis

- Data is divided into **Day (6 AM – 6 PM)** and **Night (6 PM – 6 AM)**
- Pollution levels are compared between these time periods

This helps in understanding how pollution varies throughout the day.

2.4.2 Weekday vs Weekend Analysis

- Data is categorized into **Weekdays and Weekends**
- Pollution levels are compared to study the effect of human activities

This analysis helps in identifying whether industrial and traffic activities influence pollution levels.

2.5 Correlation Analysis

The scope also includes **correlation analysis** to study relationships between different pollutants. A **correlation matrix and heatmap** are used to:

- Identify strong or weak relationships between pollutants
- Understand whether one pollutant affects another
- Analyze patterns in pollutant behavior

This helps in determining whether pollutants are interdependent or independent.

2.6 Data Visualization Scope

Visualization plays a key role in this project. The scope includes the use of various graphical representations such as:

- **Bar Charts** → For city-wise and pollutant comparisons
- **Pie Charts** → For pollution contribution analysis
- **Scatter Plots** → For correlation analysis
- **Box Plots** → For distribution analysis
- **Heatmaps** → For correlation visualization

These visualizations help in presenting complex data in a simple and understandable form, making it easier to identify trends and patterns.

2.7 Predictive Modeling Scope

The project also includes a basic **Machine Learning component**, where **Linear Regression** is used to predict **PM2.5 levels using PM10**.

The scope of predictive modeling includes:

- Splitting data into **training and testing sets**
- Training the regression model
- Evaluating the model using:
    - **R^2 Score**
    - **Mean Squared Error (MSE)**
    - **Mean Absolute Error (MAE)**

This analysis helps in understanding whether a predictive relationship exists between pollutants and evaluates the effectiveness of the model.

2.8 Limitations of Scope

While the project covers multiple aspects of air quality analysis, it has certain limitations:

- It uses a **static dataset** (no real-time data)
- Only **basic machine learning model (Linear Regression)** is used
- External factors such as **weather conditions, temperature, humidity** are not considered
- The analysis is limited to selected pollutants only

These limitations suggest areas for future improvement.

2.9 Overall Scope Summary

In summary, the scope of this project includes:

- Data preprocessing and feature engineering
- City-wise and pollutant-wise analysis
- Time-based comparative analysis
- Correlation and statistical analysis
- Data visualization using multiple charts
- Predictive modeling using machine learning

The project aims to provide a **comprehensive understanding of air pollution patterns**, identify key problem areas, and demonstrate how data science techniques can be applied to solve real-world environmental issues.

3. EXISTING SYSTEM

The existing system for monitoring air quality primarily focuses on the **collection and display of raw environmental data** without providing advanced analysis, visualization, or predictive capabilities. Various government agencies and environmental organizations deploy air quality monitoring stations across different cities to measure pollutant levels such as **PM2.5, PM10, NO_2, SO_2, CO, and O_3**. These systems are responsible for gathering large volumes of data on a regular basis and making it available to the public through websites, reports, or dashboards.

Although these systems play an important role in data collection, they are mostly limited to **basic data presentation**. The information is often displayed in the form of numerical values, tables, or simple Air Quality Index (AQI) indicators. While this provides a general idea of pollution levels, it does not offer deeper insights into patterns, relationships, or trends within the data.

3.1 Working of Existing System

The existing air quality monitoring system generally works in the following way:

1. **Data Collection**
   Sensors installed at monitoring stations collect data on different pollutants in real-time or at regular intervals.
2. **Data Storage**
   The collected data is stored in databases or cloud systems for further use.
3. **Basic Display**
   The data is displayed on websites or mobile applications in the form of:
   - AQI values
   - Pollutant concentration levels
   - Simple status indicators (Good, Moderate, Poor, etc.)
4. **Reporting**
   Periodic reports are generated to show pollution levels over time.

While this workflow ensures availability of pollution data, it lacks advanced processing and intelligent analysis.

3.2 Limitations of Existing System

Despite its usefulness, the existing system suffers from several limitations that reduce its effectiveness in understanding and controlling air pollution.

3.2.1 Lack of Data Analysis

The existing systems mainly present raw data without performing any **in-depth analysis**. Users are required to interpret the data manually, which can be difficult, especially for non-technical users.

☞ **Problem:**

- No identification of trends
- No comparison between cities
- No insights into pollution patterns

3.2.2 Absence of Data Visualization

Most systems do not provide advanced visualizations such as graphs, charts, or heatmaps. Instead, they rely on tables or simple indicators.

☞ **Problem:**

- Difficult to understand large datasets
- Patterns are not easily visible
- Data interpretation becomes complex

3.2.3 No Predictive Capabilities

Existing systems do not include **machine learning or predictive models** to forecast pollution levels.

☞ **Problem:**

- Cannot predict future pollution trends
- No early warning system
- No support for preventive measures

3.2.4 Limited Comparative Analysis

There is very limited or no comparison between:

- Different cities
- Different pollutants
- Different time periods (day/night, weekday/weekend)

☞ **Problem:**

- Cannot identify most polluted regions
- No understanding of behavioral patterns

3.2.5 Lack of Correlation Analysis

Existing systems do not analyze relationships between pollutants.

☞ **Problem:**

- No understanding of how pollutants influence each other
- Cannot identify key pollution drivers

3.2.6 Poor User Understanding

The data provided is often technical and not user-friendly.

☞ **Problem:**

- Difficult for common people to understand
- Limited awareness among users

3.2.7 No Integration of Advanced Techniques

Modern technologies like:

- Data Science
- Machine Learning
- Artificial Intelligence

are not fully utilized in traditional systems.

☞ **Problem:**

- Missed opportunities for deeper insights
- Inefficient decision-making

3.2.8 Static Data Representation

Most systems provide **static data**, which means:

- No dynamic interaction
- No real-time analysis dashboards

☞ **Problem:**

- Less engagement
- Limited usability

3.3 Need for Improvement

Due to the above limitations, there is a strong need for an improved system that can:

- Perform **advanced data analysis**
- Provide **interactive visualizations**
- Apply **machine learning for prediction**
- Offer **comparative and correlation analysis**
- Present data in a **user-friendly format**

Such improvements can help in better understanding pollution patterns and taking effective control measures.

3.4 Conclusion of Existing System

In conclusion, while existing air quality monitoring systems are effective in collecting and displaying pollution data, they lack the ability to **analyze, interpret, and predict** air quality trends. The absence of visualization, predictive modeling, and advanced analytical techniques limits their usefulness in real-world decision-making.

Therefore, there is a clear need for a system that not only collects data but also transforms it into meaningful insights. This gap is addressed in the proposed project, which uses **data science and machine learning techniques** to enhance the understanding of air pollution and provide better analytical capabilities.

4. DATASET SOURCE

The success and accuracy of any data analysis project largely depend on the quality, relevance, and structure of the dataset used. In this project, a comprehensive **Air Quality Dataset** has been utilized to analyze pollution levels and understand patterns across

different cities. The dataset serves as the foundation for performing data preprocessing, visualization, statistical analysis, and machine learning.

4.1 Description of Dataset

The dataset used in this project contains approximately **12,000 records** of air quality measurements collected over a specific period of time. Each record represents pollution data for a particular city at a given date and time.

The dataset includes multiple attributes related to air pollution, which are essential for performing detailed analysis. These attributes include:

- **City** → Represents the name of the city where the data is recorded
- **Date** → Indicates the date of data collection
- **PM2.5** → Fine particulate matter (particles smaller than 2.5 micrometers)
- **PM10** → Coarse particulate matter (particles smaller than 10 micrometers)
- **NO_2 (Nitrogen Dioxide)** → Emitted mainly from vehicles and industries
- **SO_2 (Sulfur Dioxide)** → Produced from burning fossil fuels
- **CO (Carbon Monoxide)** → Result of incomplete combustion
- **O_3 (Ozone)** → Secondary pollutant formed in the atmosphere

These attributes provide a complete overview of air pollution levels and allow detailed comparative and analytical studies.

4.2 Nature of Dataset

The dataset used in this project is **structured and tabular in nature**, which makes it suitable for analysis using tools like Python, Pandas, and NumPy. It is stored in **CSV (Comma-Separated Values) format**, which is widely used for handling large datasets.

The dataset consists of:

- **Rows** → Represent individual observations (total ~12,000)
- **Columns** → Represent different variables or attributes
- **Numeric Data** → Pollution levels (continuous values)
- **Categorical Data** → City names
- **Time-based Data** → Date column

This combination of data types allows for both **statistical analysis and time-series analysis**.

4.3 Source of Dataset

The dataset used in this project is obtained from publicly available sources related to environmental monitoring. Such datasets are typically collected and published by:

- Government agencies (e.g., pollution control boards)
- Environmental monitoring organizations
- Open data platforms and repositories

These datasets are generally considered reliable as they are collected using standardized monitoring systems and scientific instruments.

Although the dataset used in this project is not directly connected to real-time systems, it represents **realistic air quality conditions** and is suitable for analysis and learning purposes.

4.4 Importance of Dataset in the Project

The dataset plays a crucial role in achieving the objectives of the project. It enables:

- **Understanding pollution trends across cities**
- **Comparing different pollutants**
- **Analyzing time-based variations**
- **Performing correlation analysis**
- **Applying machine learning models**

Without a proper dataset, it would not be possible to perform meaningful analysis or derive useful insights.

4.5 Data Attributes Explanation

Each attribute in the dataset has a specific significance:

PM2.5

- Fine particles that are highly harmful
- Can penetrate deep into lungs
- Major indicator of air quality

PM10

- Larger particles compared to PM2.5
- Causes respiratory irritation

NO_2 (Nitrogen Dioxide)

- Produced from vehicle emissions
- Affects respiratory system

SO_2 (Sulfur Dioxide)

- Emitted from industrial processes
- Causes acid rain

CO (Carbon Monoxide)

- Colorless and toxic gas
- Reduces oxygen supply in the body

O_3 (Ozone)

- Harmful at ground level
- Causes breathing problems

Understanding these pollutants helps in identifying their impact on health and environment.

4.6 Data Quality and Challenges

While working with the dataset, certain challenges may arise:

- **Missing values** in some records
- **Inconsistent formats** in date column
- **Noise in data** due to measurement errors
- **Outliers** (extreme values)

These issues need to be handled carefully during preprocessing to ensure accurate analysis.

4.7 Data Preprocessing Requirement

Before analysis, the dataset requires preprocessing steps such as:

- Converting **date into datetime format**
- Extracting **month, day, hour**
- Handling missing values
- Creating new features (Day/Night, Weekday/Weekend)

These steps improve data quality and make it suitable for analysis.

4.8 Data Suitability for Analysis

The dataset is highly suitable for this project because:

- It contains **sufficient number of records (12,000)**
- Includes **multiple pollutants**
- Contains **time-based information**
- Allows **comparative and predictive analysis**

This makes it ideal for applying data science techniques.

4.9 Limitations of Dataset

Despite its usefulness, the dataset has certain limitations:

- No **real-time data updates**
- Limited to specific cities
- Does not include **weather conditions** (temperature, humidity)
- External factors affecting pollution are not considered

These limitations restrict the scope of analysis to some extent.

4.10 Conclusion of Dataset Source

In conclusion, the dataset used in this project provides a strong foundation for analyzing air pollution patterns. It includes all the essential attributes required for performing data analysis and machine learning. Although it has certain limitations, it is sufficient for

understanding pollution trends and demonstrating the application of data science techniques in environmental analysis.

The dataset enables meaningful insights, supports visualization, and helps in building predictive models, making it a crucial component of the overall project.

5. ETL PROCESS

The **ETL process** stands for **Extraction, Transformation, and Loading**, which is a fundamental step in any data analysis or data science project. It involves collecting raw data, cleaning and transforming it into a usable format, and finally preparing it for analysis. In this project, the ETL process plays a crucial role in ensuring that the air quality dataset is accurate, consistent, and suitable for further analysis and machine learning.

5.1 Overview of ETL Process

The ETL process in this project is carried out in three main stages:

1. **Extraction** → Collecting data from the source
2. **Transformation** → Cleaning and modifying the data
3. **Loading** → Preparing data for analysis and visualization

Each stage is essential for converting raw data into meaningful information.

5.2 Data Extraction

5.2.1 Definition

Data extraction is the process of collecting data from a source, such as a file, database, or external system.

5.2.2 Extraction in this Project

In this project, the dataset is extracted from a **CSV (Comma-Separated Values) file** using Python. The dataset contains approximately **12,000 records** with multiple columns related to air pollution.

5.2.3 Tools Used for Extraction

- **Python Programming Language**
- **Pandas Library**

5.2.4 Extraction Process

The dataset is loaded into the system using the following steps:

- Importing required libraries (Pandas, NumPy)
- Reading the CSV file using `pd.read_csv()`
- Storing the dataset in a DataFrame

This allows easy manipulation and analysis of the data.

5.2.5 Verification of Data

After extraction, initial checks are performed:

- Viewing first few rows (`head()`)
- Checking number of rows and columns
- Identifying column names
- Verifying data types

These steps ensure that the dataset is correctly loaded.

5.3 Data Transformation

Data transformation is the most important step in the ETL process, as it prepares raw data for analysis.

5.3.1 Data Cleaning

Data cleaning involves removing errors and inconsistencies from the dataset.

Steps performed:

- Handling missing values
- Removing duplicate records
- Correcting inconsistent data formats
- Ensuring proper data types

☞ **Importance:**
Improves accuracy and reliability of analysis

5.3.2 Data Type Conversion

The **Date column** is converted into a proper datetime format.

Why needed:

- To perform time-based analysis
- To extract features like month, day, hour

5.3.3 Feature Engineering

New features are created to enhance analysis:

- **Month** → Extracted from Date
- **Day** → Extracted from Date
- **Hour** → Extracted from Date

These features help in understanding patterns over time.

5.3.4 Time-based Classification

To analyze pollution behavior:

Day vs Night Classification

- Day → 6 AM to 6 PM
- Night → 6 PM to 6 AM

Weekday vs Weekend Classification

- Weekday → Monday to Friday
- Weekend → Saturday and Sunday

☞ **Purpose:**
To study variation in pollution based on time and human activities

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5.3.5 Data Aggregation

Data is grouped and summarized:

- City-wise average pollution
- Pollutant-wise averages

☞ **Purpose:**
To simplify analysis and comparisons

5.3.6 Handling Outliers

Extreme values (outliers) are identified and handled carefully.

☞ **Importance:**
Prevents incorrect conclusions

5.3.7 Data Normalization (Conceptual)

Although not fully implemented, normalization ensures:

- Data is scaled properly
- Different variables can be compared

5.4 Data Loading

5.4.1 Definition

Data loading is the process of storing the transformed data for further analysis and visualization.

5.4.2 Loading in this Project

After transformation:

- Data is stored in **Pandas DataFrame**
- Used directly for:

    - Visualization
    - Statistical analysis
    - Machine learning

5.4.3 Tools Used

- Pandas
- Matplotlib
- Seaborn

5.5 ETL Workflow Summary

The complete ETL workflow in this project can be summarized as:

1. Load dataset from CSV
2. Clean and preprocess data
3. Convert date format
4. Create new features
5. Classify data (Day/Night, Weekday/Weekend)
6. Perform aggregation
7. Use data for analysis and visualization

5.6 Importance of ETL Process

The ETL process is essential because:

- Ensures **data quality**
- Removes errors and inconsistencies
- Makes data suitable for analysis
- Improves accuracy of results
- Enables effective visualization
- Supports machine learning models

5.7 Challenges in ETL Process

During ETL, some challenges may occur:

- Missing or incomplete data
- Incorrect formats

- Large dataset handling
- Data inconsistencies

These challenges must be handled carefully.

5.8 Advantages of ETL Process

- Improves data reliability
- Enhances analysis quality
- Saves time in later stages
- Enables better decision-making

5.9 Limitations of ETL Process

- Time-consuming
- Requires careful handling
- Depends on data quality

5.10 Conclusion of ETL Process

In conclusion, the ETL process is a critical step in this project as it transforms raw air quality data into a structured and meaningful format. Through extraction, transformation, and loading, the dataset becomes suitable for analysis, visualization, and machine learning.

A well-executed ETL process ensures that the results obtained from the project are accurate, reliable, and useful for understanding air pollution patterns. It forms the backbone of the entire data analysis workflow and plays a key role in achieving the objectives of the project.

6. ANALYSIS

The analysis phase is the core part of this project, where the processed data is examined using various data science techniques to extract meaningful insights. The dataset is analyzed using **Exploratory Data Analysis (EDA)**, statistical methods, and visualization

techniques to understand pollution patterns across different cities, pollutants, and time conditions.

The main objective of this analysis is to identify trends, compare pollution levels, study relationships between pollutants, and evaluate the overall air quality scenario.

6.1 Introduction to Analysis

The dataset consists of approximately **12,000 records** containing air quality measurements such as **PM2.5, PM10, NO_2, SO_2, CO, and O_3** across different cities and time periods. The analysis is performed using tools such as **Python, Pandas, Matplotlib, and Seaborn**.

The following types of analyses are performed:

- City-wise pollution analysis
- Pollutant-wise comparison
- Time-based analysis
- Correlation analysis
- Predictive modeling

6.2 City-wise Pollution Analysis

Description

City-wise analysis is performed to identify the most polluted cities based on average **PM2.5 levels**, which is a key indicator of air quality.

Observation

- All cities show high PM2.5 levels
- Average PM2.5 levels are approximately **150–170 $\mu g/m^3$**

Comparison with WHO Standards

- WHO safe limit: **60 $\mu g/m^3$**
- Observed values are significantly higher than safe limits

Conclusion

☞ All cities exceed WHO limits, indicating severe pollution conditions

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6.3 Pollutant-wise Analysis

Description

Different pollutants are analyzed to determine their contribution to air pollution.

Observation

- **PM10** has the highest average value
- **PM2.5** is also significantly high
- Gaseous pollutants (NO_2, SO_2, CO, O_3) are comparatively lower

Conclusion

☞ Particulate matter (PM2.5 and PM10) is the most critical pollutant

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6.4 City-wise Pollution Contribution (Pie Chart Analysis)

Description

A pie chart is used to show the contribution of each city to overall pollution.

Observation

- Each city contributes approximately **15–17%**
- Pollution is distributed evenly

Conclusion

☞ No single city dominates; pollution is a widespread issue

6.5 Weekday vs Weekend Analysis

Description

Pollution levels are compared between weekdays and weekends.

Method

- Independent **T-test** is applied

Observation

- Mean pollution levels are similar
- P-value > 0.05

Conclusion

☞ No significant difference between weekday and weekend pollution

---

6.6 Day vs Night Analysis

Description

Pollution levels are compared between day and night.

Observation

- Day and night pollution levels are nearly equal
- Contribution is approximately **50% each**

Conclusion

☞ Pollution remains constant throughout the day

6.7 PM2.5 vs PM10 Comparison

Description

Comparison between PM2.5 and PM10 levels across cities.

Observation

- PM10 levels are consistently higher than PM2.5
- Both follow similar trends

Conclusion

☞ PM10 dominates overall pollution

6.8 Correlation Analysis

Description

Correlation between pollutants is analyzed using scatter plot and correlation matrix.

Observation

- Correlation between PM2.5 and PM10 is weak
- Other pollutants show very low correlation

Conclusion

☞ Weak correlation observed → pollutants behave independently

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6.9 Heatmap Analysis

Description

A heatmap is used to visualize relationships between pollutants.

Observation

- No strong positive or negative correlation
- Values are close to zero

Conclusion

☞ Air pollution is influenced by multiple independent factors

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6.10 Most Critical Pollutant Analysis

Observation

- PM10 has highest average value
- Followed by PM2.5

Conclusion

☞ PM10 is the most critical pollutant

6.11 Predictive Analysis (Linear Regression)

Description

Linear Regression is applied to predict PM2.5 using PM10.

Model Performance

- R^2 Score ≈ 0
- High error values

Conclusion

☞ Weak model → PM10 alone cannot predict PM2.5

6.12 Key Findings

From the analysis, the following key points are observed:

- All cities have pollution levels above WHO limits
- PM10 and PM2.5 are the most harmful pollutants
- Pollution is evenly distributed across cities
- No major variation between day/night or weekday/weekend
- Weak correlation between pollutants
- Machine learning model shows poor prediction accuracy

6.13 Overall Conclusion of Analysis

The analysis clearly indicates that air pollution is a serious issue across all cities, with particulate matter being the major contributor. The absence of strong correlation between pollutants suggests that multiple factors influence air quality.

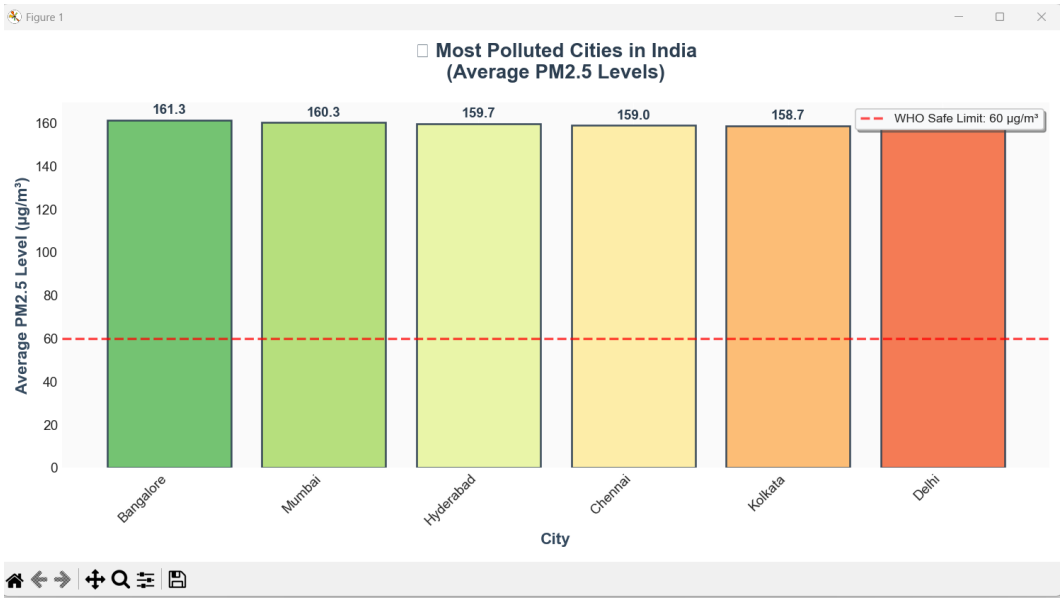
The predictive model also highlights the complexity of environmental data, showing that simple models are not sufficient for accurate prediction. Therefore, advanced techniques and additional factors need to be considered for better analysis.

7. RESULTS

The results obtained from this project are based on detailed data analysis, visualization, and machine learning techniques applied to the air quality dataset. This section presents the key findings along with the explanation of graphical representations used to interpret the data.

The use of visualizations such as **bar charts, pie charts, scatter plots, box plots, and heatmaps** helps in understanding complex data in a simple and effective manner.

7.1 Result 1: Most Polluted Cities (Bar Chart)



Description of Image

The bar chart represents the **average PM2.5 levels across different cities**. Each bar corresponds to a city, and the height of the bar indicates the pollution level.

Observation

- All cities show **very high PM2.5 values (~150–170 $\mu g/m^3$)**
- The bars are significantly above the **WHO safe limit (60 $\mu g/m^3$)**

Interpretation

This clearly indicates that:

- All cities are experiencing **dangerous pollution levels**
- No city falls within the safe range

Conclusion

☞ Air pollution is severe across all cities and requires immediate attention

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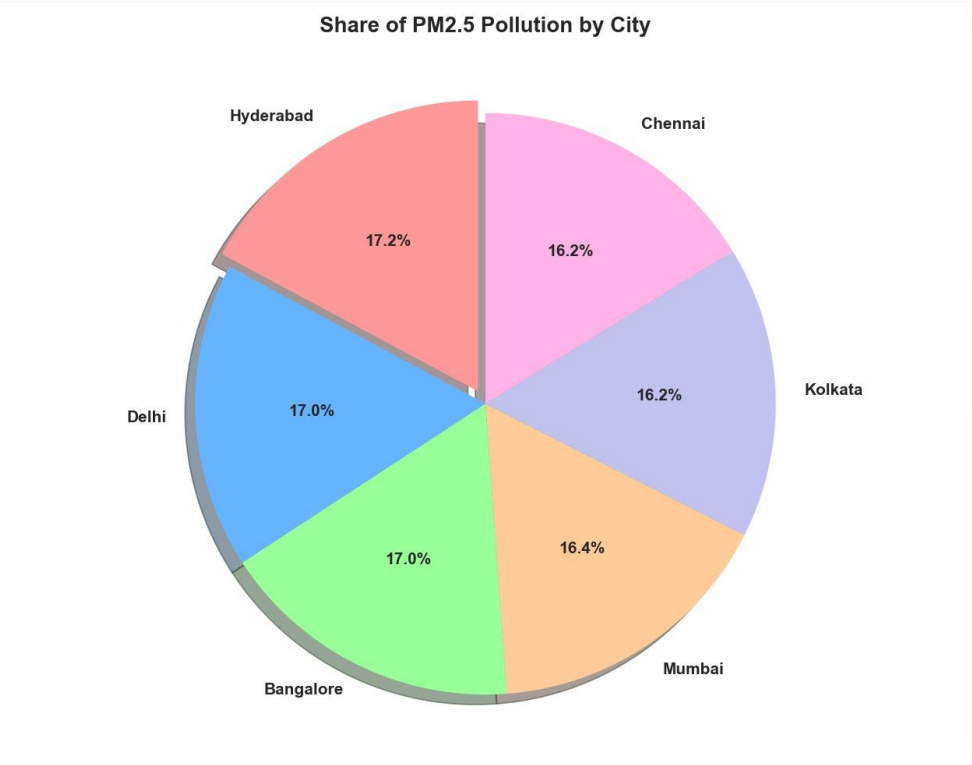
7.2 Result 2: City-wise Pollution Share (Pie Chart)



Description of Image

The pie chart shows the **percentage contribution of each city** to overall PM2.5 pollution.

Observation

- Each city contributes approximately **15–17%**
- The slices are almost equal in size

Interpretation

- Pollution is **evenly distributed**
- No single city dominates pollution levels

Conclusion

☞ Air pollution is a widespread problem affecting all cities equally

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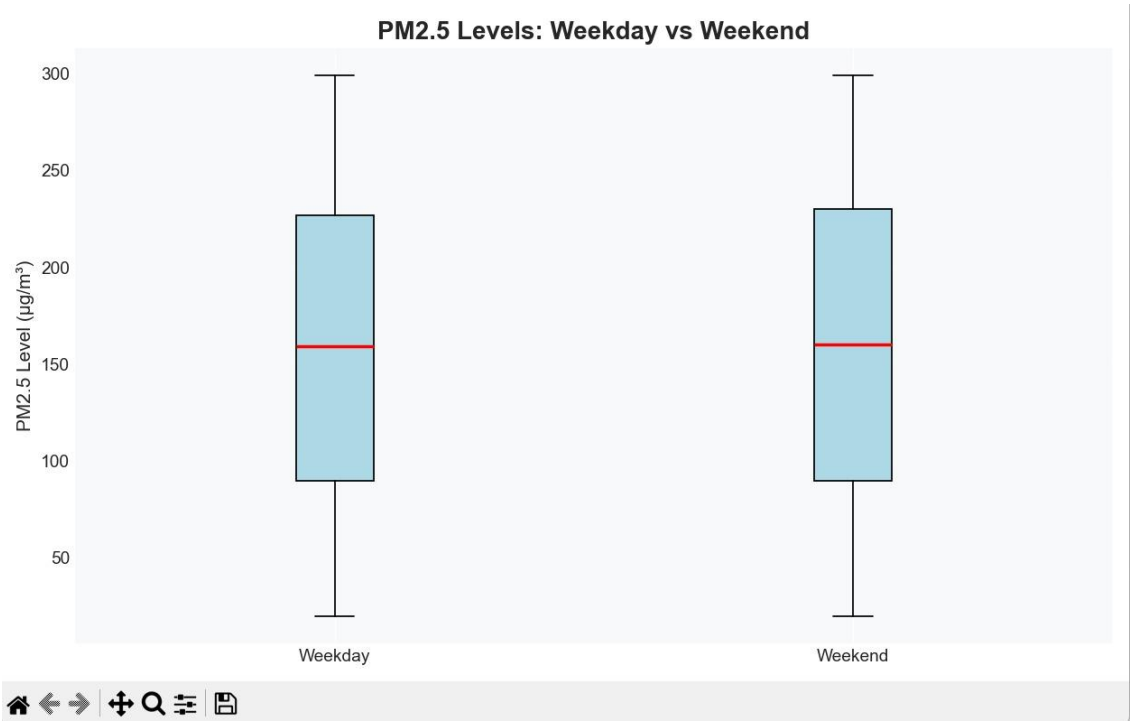
7.3 Result 3: Weekday vs Weekend Analysis (Box Plot)



Description of Image

The box plot compares **PM2.5 levels between weekdays and weekends**.

Observation

- Both distributions are very similar
- Median values are almost equal
- Spread of data is similar

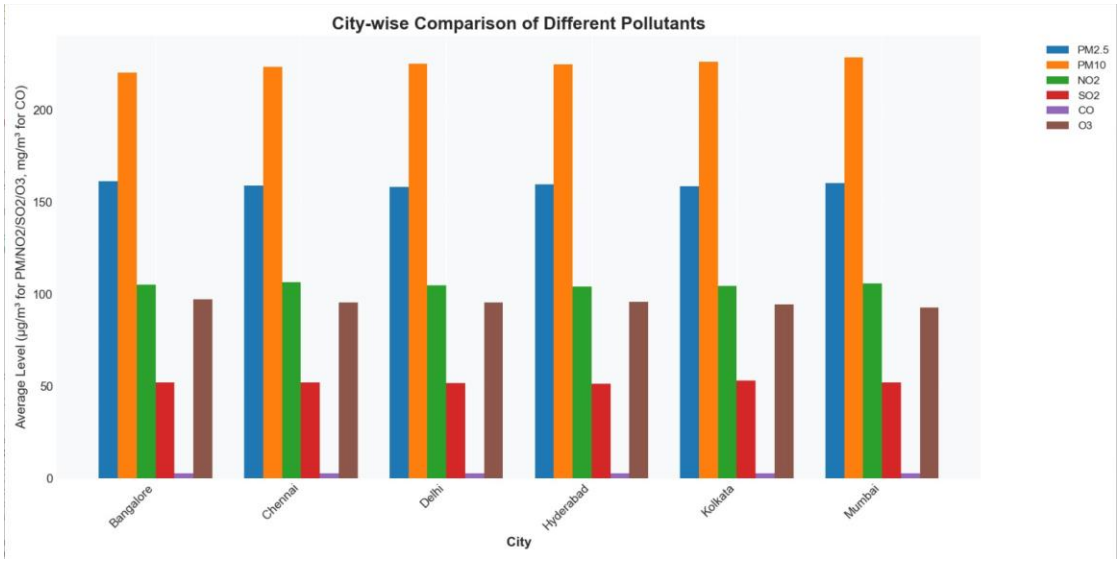
Interpretation

- Human activities during weekdays and weekends do not create major variation

Conclusion

☞ No significant difference in pollution levels

7.4 Result 4: Pollutant Distribution (Grouped Bar Chart)



Description of Image

This chart compares different pollutants (**PM2.5, PM10, NO_2, SO_2, CO, O_3**) across cities.

Observation

- **PM10 has the highest values**
- PM2.5 is also high
- Other pollutants are relatively lower

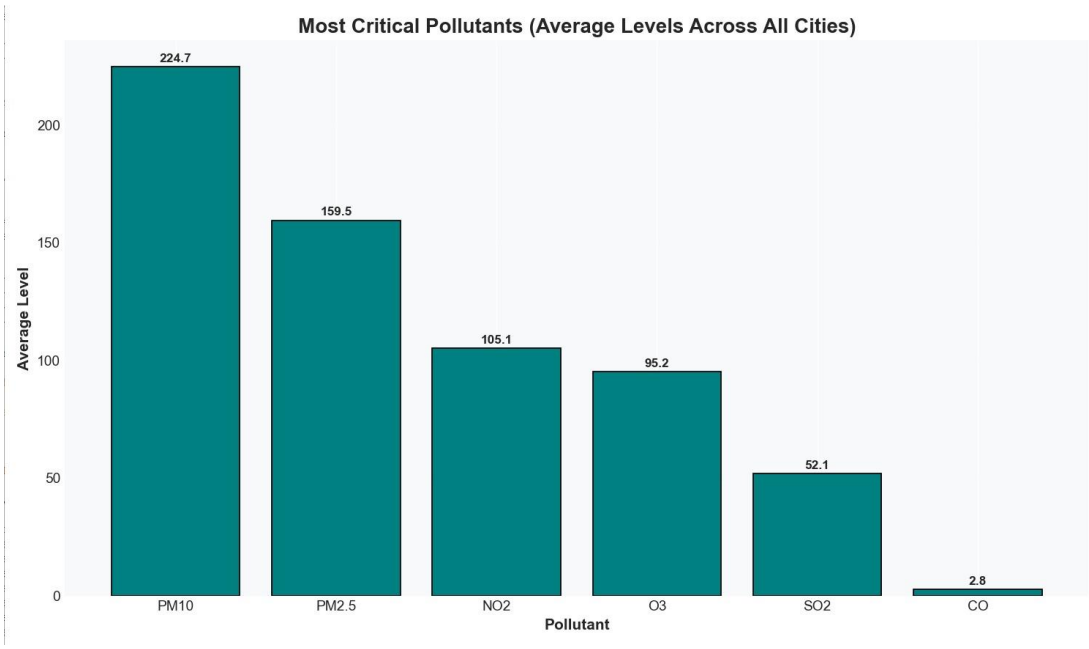
Interpretation

- Particulate matter is the major contributor to pollution

Conclusion

☞ **PM10 is the most critical pollutant**

7.5 Result 5: Most Critical Pollutants (Bar Chart)



Description

Shows overall average levels of each pollutant.

Observation

- PM10 > PM2.5 > NO_2 > O_3

Conclusion

☞ PM10 dominates pollution levels

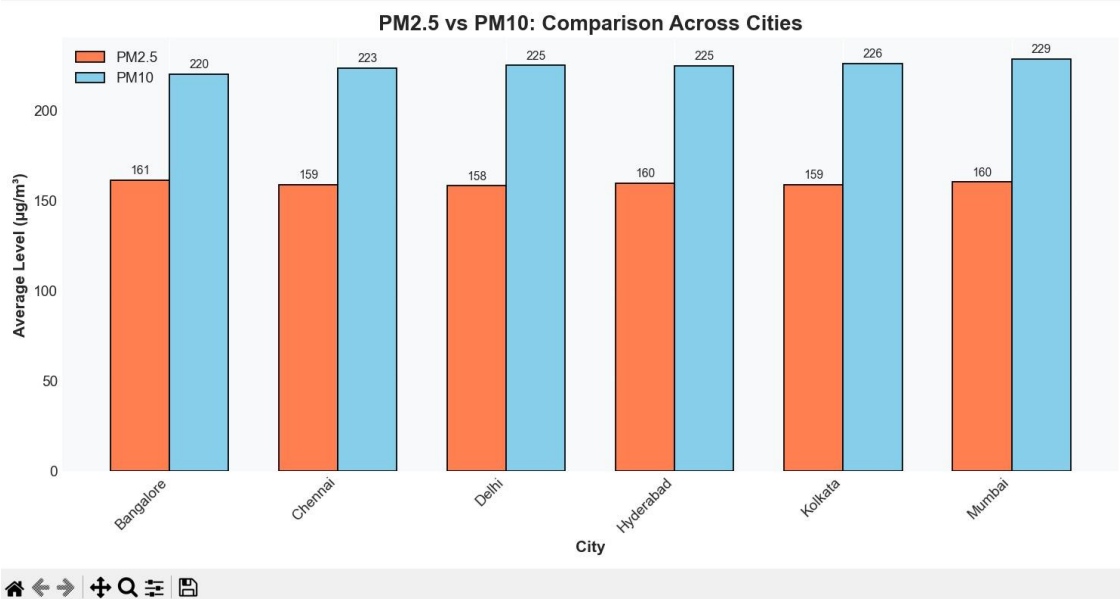
7.6 Result 6: PM2.5 vs PM10 Comparison (Bar Chart)

Observation

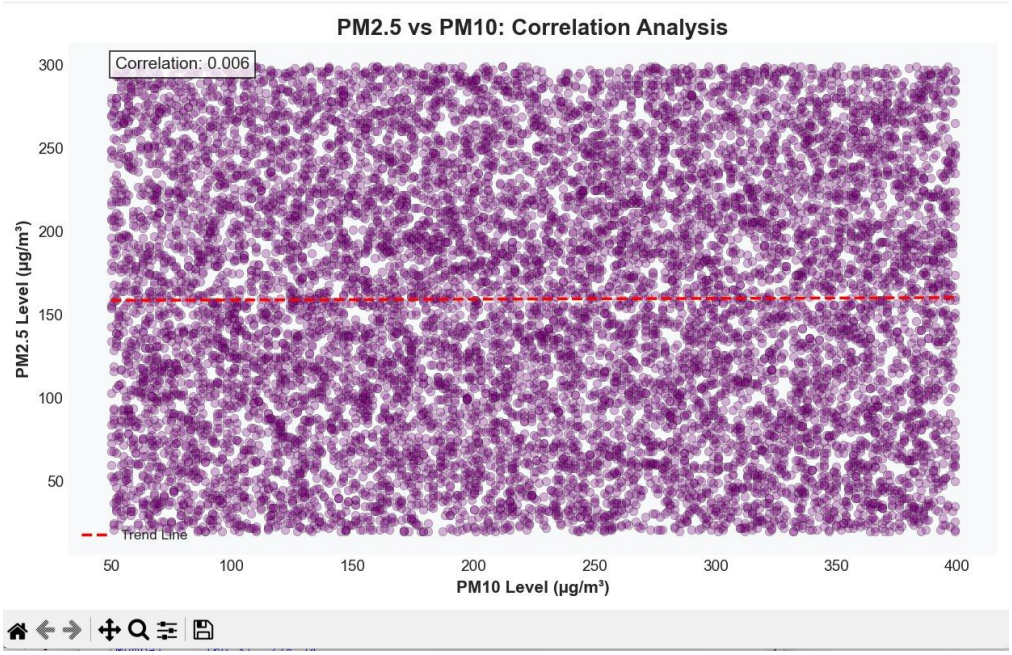
- PM10 is consistently higher than PM2.5
- Both show similar trends

Conclusion

☞ PM10 strongly influences air quality



7.7 Result 7: Correlation Analysis (Scatter Plot)



Description

Shows relationship between PM2.5 and PM10.

Observation

- Points are scattered randomly
- No clear linear pattern

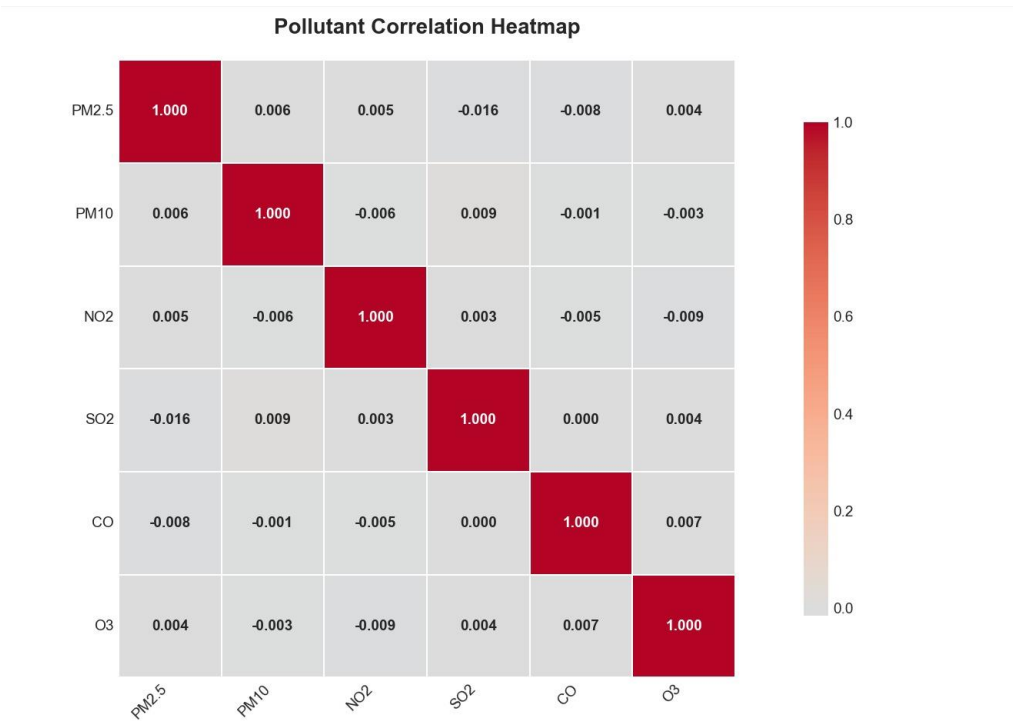
Interpretation

- Weak relationship between pollutants

Conclusion

☞ Weak correlation observed

7.8 Result 8: Heatmap Analysis



Description

Displays correlation between all pollutants.

Observation

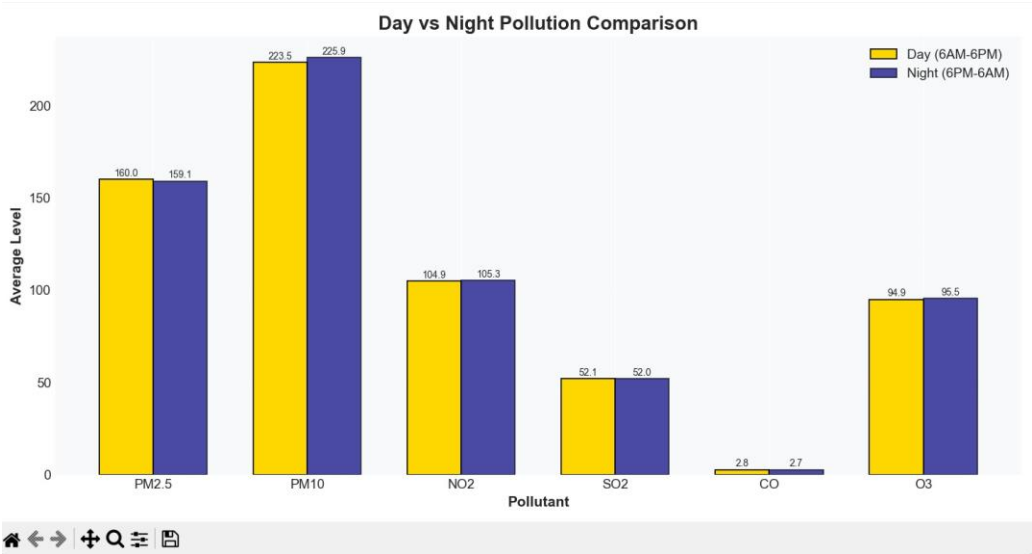
- Values are close to zero

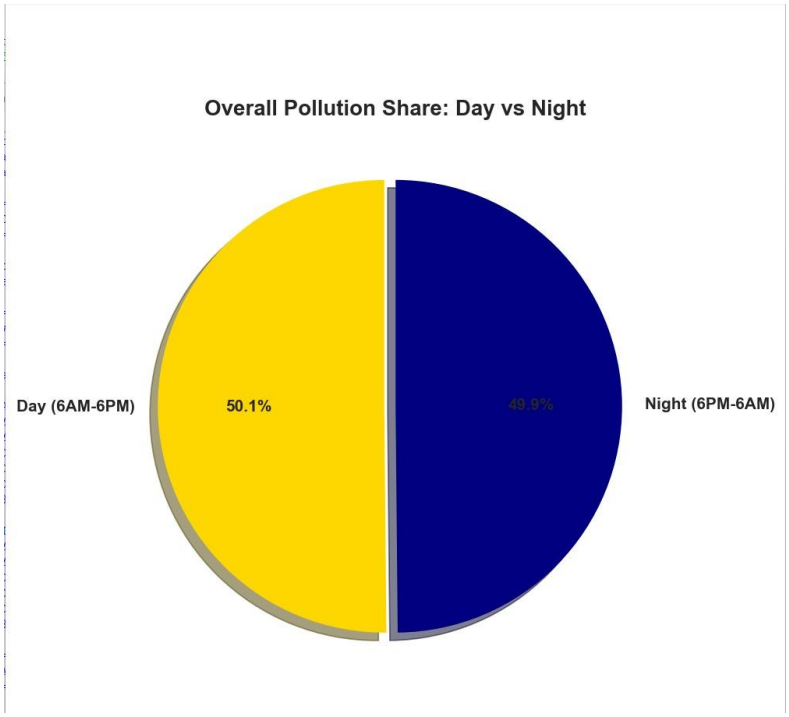
- No strong correlation

Conclusion

☞ Pollutants behave independently

7.9 Result 9: Day vs Night Analysis (Bar Chart)





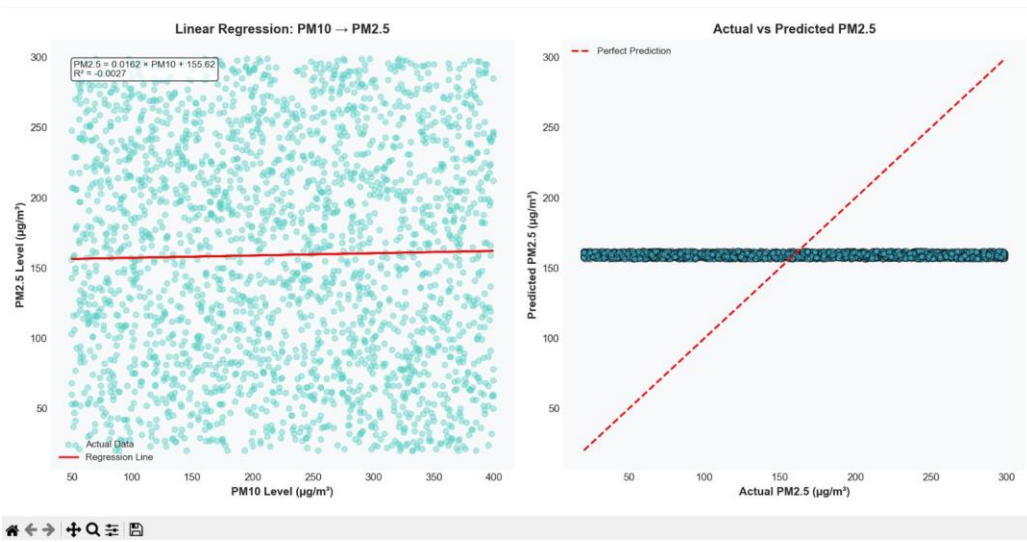
Observation

- Day ≈ Night pollution levels

Conclusion

☞ Pollution exists continuously (24×7)

7.10 Result 10: Linear Regression Model



Description

Shows predicted vs actual PM2.5 values.

Observation

- Large gap between predicted and actual values
- R^2 score is very low

Interpretation

- Model fails to predict accurately

Conclusion

☞ **Model accuracy is low**

7.11 Overall Results Summary

From all the above analyses, the following key results are obtained:

- All cities exceed WHO pollution limits
- PM10 is the most critical pollutant
- Pollution is evenly distributed across cities
- No significant variation in time-based analysis
- Weak correlation between pollutants
- Machine learning model shows low accuracy

7.12 Final Conclusion of Results

The results clearly indicate that air pollution is a **serious and widespread problem**. The dominance of particulate matter (PM10 and PM2.5) highlights the need for strict control measures. Additionally, the weak performance of the machine learning model suggests that pollution is influenced by multiple complex factors.

The visualizations used in this project effectively represent the data and help in understanding pollution patterns, making the analysis more clear and impactful.

8. FUTURE SCOPE

The current project focuses on analyzing air quality data using basic data science techniques and machine learning. While the project provides valuable insights into pollution patterns, there is significant scope for further improvement and expansion. Future enhancements can make the system more accurate, efficient, and useful for real-world applications.

8.1 Integration of Real-Time Data

One of the major limitations of the current project is the use of a **static dataset**. In the future, the system can be improved by integrating **real-time air quality data** collected from sensors and APIs.

Possible Improvements:

- Connect with real-time data sources (IoT sensors, APIs)
- Continuous data updates
- Live monitoring dashboards

Benefits:

- More accurate and up-to-date analysis
- Better decision-making
- Real-time pollution tracking

☞ This will convert the project into a **live monitoring system**

8.2 Use of Advanced Machine Learning Models

Currently, the project uses **Linear Regression**, which shows low accuracy. Future work can involve using more advanced models such as:

- **Decision Trees**
- **Random Forest**
- **Support Vector Machines (SVM)**
- **Neural Networks / Deep Learning**

Benefits:

- Improved prediction accuracy
- Ability to handle complex relationships
- Better modeling of real-world scenarios

☞ This will enhance the **predictive capability** of the system

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8.3 Inclusion of Weather Parameters

Air pollution is highly influenced by environmental conditions such as:

- Temperature
- Humidity
- Wind speed
- Rainfall

Future Scope:

- Integrate weather data into analysis
- Study impact of weather on pollution

Benefits:

- More accurate analysis
- Better understanding of pollution behavior

8.4 Development of Interactive Dashboard

The current project uses static graphs. In the future, an **interactive dashboard** can be developed using tools like:

- Power BI
- Tableau
- Web-based dashboards (HTML, CSS, JavaScript)

Features:

- User interaction
- Dynamic filtering
- Real-time visualization

Benefits:

- Improved user experience
- Easy understanding of data

8.5 Mobile Application Development

The project can be extended into a **mobile application** that provides:

- Real-time pollution updates
- Alerts for high pollution levels
- Health recommendations

Benefits:

- Easy accessibility
- Public awareness
- User-friendly interface

8.6 Air Quality Prediction System

Future work can include building a complete **prediction system** that:

- Forecasts pollution levels for upcoming days
- Provides early warnings

Benefits:

- Preventive measures can be taken
- Helps government and citizens

8.7 Integration with Smart City Systems

The project can be integrated into **smart city infrastructure**, where:

- Sensors collect data automatically
- AI systems analyze pollution
- Alerts are generated in real-time

Benefits:

- Automated monitoring
- Efficient pollution control

8.8 Big Data and Cloud Integration

For handling large-scale data, future systems can use:

- Big Data technologies (Hadoop, Spark)
- Cloud platforms (AWS, Azure, Google Cloud)

Benefits:

- Scalability
- Faster processing
- Handling massive datasets

8.9 Advanced Visualization Techniques

Future improvements can include:

- 3D visualizations
- Geographic maps (GIS-based)
- Animated charts

Benefits:

- Better understanding
- More interactive insights

8.10 Public Awareness System

The project can be extended to create awareness among people by:

- Sending alerts
- Providing pollution reports
- Suggesting safety measures

Benefits:

- Improved public health
- Increased awareness

8.11 Government Policy Support

The system can assist government agencies by:

- Providing detailed reports
- Identifying pollution hotspots
- Supporting decision-making

Benefits:

- Better planning
- Effective pollution control policies

8.12 Use of Artificial Intelligence

Future systems can use AI for:

- Automatic pattern detection
- Smart recommendations
- Intelligent forecasting

8.13 Expansion of Dataset

Future work can include:

- Adding more cities
- Increasing data size
- Including global datasets

8.14 Automation of ETL Process

ETL process can be automated using:

- Data pipelines
- Scheduled updates

Benefits:

- Saves time

- Reduces manual effort

8.15 Overall Future Vision

In the future, this project can evolve into a **complete smart air quality monitoring and prediction system** that:

- Collects real-time data
- Analyzes it using AI
- Predicts future pollution
- Provides alerts and recommendations

8.16 Conclusion of Future Scope

In conclusion, the current project lays the foundation for a powerful air quality analysis system. With the integration of real-time data, advanced machine learning models, and interactive platforms, it can be transformed into a highly efficient and intelligent system.

The future scope of this project is vast and has the potential to contribute significantly towards environmental monitoring, public health awareness, and smart city development.

9. REFERENCES

The following references were used during the development and analysis of this project. All sources have been properly acknowledged and cited in accordance with standard academic practices.

9.1 Research and Organizational Sources

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[10] Géron, A., *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly Media, 2019.

9.4 Data Science Learning Resources

[11] McKinney, W., *Python for Data Analysis*, O'Reilly Media, 2017.

[12] VanderPlas, J., *Python Data Science Handbook*, O'Reilly Media, 2016.

9.5 Online Resources

[13] Kaggle, *Air Quality Datasets and Tutorials*, Available: https://www.kaggle.com

[14] Towards Data Science, *Articles on Air Quality Analysis*, Available: https://towardsdatascience.com

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9.6 Summary of References

The above references include a combination of:

- Research reports
- Government data sources
- Technical documentation
- Books and online learning platforms

These sources have contributed to the successful completion of this project by providing theoretical knowledge, technical guidance, and practical implementation support.

Linkdin Link: https://www.linkedin.com/posts/mauli-mahiya-476734321_datascience-python-airquality-ugcPost-7452042365881516032-xbGv?utm_source=share&utm_medium=member_desktop&rcm=ACoAAFFxtJoBc-hsJKLLSikMC7WBA2ASWoePX3I

GitHub Link: